

ADITYA ANANTHULA

+91 9703456037 ♦ Pune, IN ♦ aditya.ind8905@gmail.com ♦ linkedin.com/in/aditya-ananthula

EDUCATION

Manipal Institute of Technology, Bengaluru
Bachelor of Technology in Computer Science (AI)

Bengaluru, Karnataka, IN
July 2023 – May 2027

FIITJEE Junior College, Miyapur
Higher Secondary School

Hyderabad, IN
Aug. 2021 – May 2023

EXPERIENCE

Software Engineering Intern
Persistent Systems, Enterprise & IT Applications, PiAssist Team

May 2026 – Present (Expected Nov 2026)
Pune, IN

- Shipped **5+ production-grade features** for an enterprise Microsoft Teams **AI chatbot**, building the **Python** backend logic that integrates LLM-driven responses with live **REST API** data sources.
- Owned the reliability of the agent's core task-processing pipeline, **evaluating and tuning AI response quality** across releases and tracing failures to their root cause in **Linux**-based environments.
- Diagnosed and resolved **3+ production defects** in the chatbot's AI response logic through systematic **benchmarking and error analysis**, improving output accuracy and release stability.

Teaching Intern
Blue Blocks International School

May 2025 – July 2025
Hyderabad, IN

- Designed and delivered a full astronomy curriculum aligned with Cambridge standards for **100+ students**, translating complex technical concepts into structured, age-appropriate lessons.
- Applied classroom management strategies that measurably improved student engagement, sustaining a positive learning environment across multiple grade levels.

PROJECTS

LoRAForge — LLM Fine-Tuning & Evaluation Pipeline | Python, PyTorch, Hugging Face Transformers, PEFT, LoRA/QLoRA, FastAPI, Docker

- Fine-tuning **open-source LLMs** (Mistral 7B, LLaMA, Gemma) on domain-specific datasets using **LoRA and QLoRA** via **Hugging Face PEFT**, with 4-bit quantization for single-GPU training.
- Building **data preprocessing and curation pipelines** (JSONL dataset prep, tokenization, train/val splits) in **pandas** to support reproducible model training workflows.
- Developing a **model evaluation and benchmarking harness** comparing base vs. fine-tuned checkpoints on accuracy and latency, with experiment tracking in **Weights & Biases**.
- Serving fine-tuned checkpoints through a **scalable FastAPI REST API**, containerized with **Docker** for reproducible deployment.

PromptBench — Prompt Engineering & Inference Optimization Suite | Python, PyTorch, Transformers, vLLM, FastAPI, React, Docker

- Building a **prompt engineering and evaluation platform** that benchmarks prompt variants across multiple open-source LLMs, scoring outputs for accuracy, consistency, and hallucination rate.
- Implementing **inference optimization** experiments comparing quantization strategies and **vLLM**-based serving to measure throughput and latency improvements.
- Exposing benchmarking results through a **FastAPI** backend and a **React** dashboard, enabling side-by-side comparison of model and prompt performance.
- Applying **NLP fundamentals** (transformer attention, tokenization strategies, context-window management) to analyze and explain model behaviour across test cases.

SKILLS

Languages: Python, C/C++, Java, SQL, JavaScript/TypeScript, HTML/CSS

AI/ML & LLMs: PyTorch, TensorFlow, Hugging Face Transformers, PEFT, LoRA/QLoRA, Fine-Tuning, Prompt Engineering, Model Evaluation, NLP (Transformers, Tokenization, Attention), Quantization, vLLM

Backend & APIs: FastAPI, Flask, RESTful API Design, Uvicorn, SQLAlchemy (ORM), PostgreSQL, MySQL

Data & MLOps: pandas, NumPy, Matplotlib, JSON/JSONL Dataset Curation, Weights & Biases, Jupyter Notebook

Tools & Platforms: Git/GitHub, Linux/Unix, Docker, Google Cloud Platform, Azure DevOps, VS Code, PyCharm